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(57) Abstract :
It is an invention in the domain of safe online communication and cryptographic information processing, specifically in meeting the requirement of safeguarding against quantum attacks in the contemporary data exchange networks. It offers a flexible cryptographic accelerator that is intended to execute classic and post-quantum algorithms on a single chip high-performance architecture. The system consists of a hybrid cryptographic processing core that supports both classical and post-quantum modules, of a parallel arithmetic core to execute computationally intensive functions, and a memory management layer to support large key structures and high capacity data movement. A hybrid security controller integrates classical, post-quantum, and hybrid execution modes and a configuration and algorithm abstraction layer allows to flexibly select algorithms as well as allow the integration of the changing post-quantum standards in the future. I/O integration interface enables a smooth deployment across embedded systems, cloud computing systems, as well as high-performance computing systems. The invention allows to execute cryptographic operations faster to beat the latency, throughput constraints and computing costs of software based implementations of post-quantum algorithms. It offers flexibility to support a variety of cryptographic families, scalability to work across a variety of operational settings and efficiency improvements to minimize power usage and resource loads. The invention will facilitate a robust migration to quantum infrastructures by integrating classical and post-quantum security systems. In general, it provides a future-proof and versatile platform to facilitate secure and high-performance data transactions due to emerging threats of quantum-computing.

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